Sensorimotor feedback for interactive realism: evaluation of a haptic driving paradigm for a forklift simulator

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Abstract. The Virtual Environments are more and more applied in industry to learn the manipulation of some complex and/or dangerous equipments. In this context, we are developing a virtual simulator dedicated for learning the manipulation of a specific forklift. In real use, this kind of forklift requires a physical involvement of people to manipulate it. This paper proposes thus an innovative haptic paradigm for the virtual driving of such a forklift. Within the virtual simulation, it aims to provide a sensorimotor stimulation which is close to the one that users have on the real forklift. Taking in account the physical behaviours of the real forklift (inertia, damping, turning radius), we haptically simulate the mechanical hinge constraints of the forklift handle to control the forklift direction. In addition, a specific haptic push/pull technique was designed to control the forklift velocity. Our hypothesis is that our paradigm is more realistic compared to classical interactive techniques, such as joystick without force feedback. To evaluate this realism, we conducted an ergonomic study of a driving task. The same driving task has been performed in three conditions: in large room with the real forklift (R), and in a Virtual Environment with a joystick (J), and with our haptic paradigm (H). We conclude that our hypothesis is verified when revisiting different acceptations of the realism concept in Virtual Environment, such as performance transfers, but also the behaviour and psychological processes transfers.

Keywords: Virtual Reality, Haptic, Driving simulation, Realism.

1 Introduction

Virtual Reality (VR) and immersive technologies provide solutions for simulating realistic situations in a controlled and secure context. That is especially required for learning applied to industry field, for instance when novice users are starting to learn the use of complex and dangerous equipments. In this context, we designed an interactive paradigm for the driving of a forklift in a virtual simulator. Our hypothesis is that more the interactive paradigm is realistic, more the transfer skills between virtual and real conditions is efficient [1]. However, the dimensions of the realism are many. The most studied ones are relating to the visual immersion, to the affordance of the virtual objects, and to the involvement of the users in the immersive task [2].

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In this paper we mainly question the role of sensorimotor stimulations for the realism of the interaction. This channel is crucial in tasks that require physical involvement of the subject. That is exactly the case for driving our targeted forklift, because it is a non-motorised vehicle. Indeed, the forklift is controlled by manipulation of its handle to push/pull it, and control its direction (Figure 1). Actually, haptic technologies are supposed to provide solutions to stimulate the sensorimotor channels of the users. However, haptic interactions have some limitations in terms of workspace, precision, range of the movements. We explain how we overcome these drawbacks to design haptic paradigm taking in account the mechanical specificities of the real forklift, to provide a realistic driving control on the forklift in virtual context, in terms of physical engagement of the user and perception of the handle.

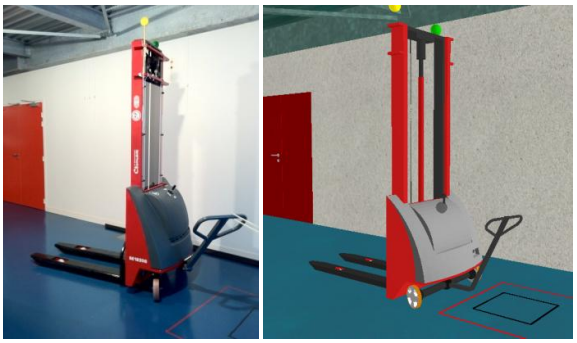


Fig. 1. The real and virtual non-motorised forklift

To evaluate the realism of our paradigm, we compare it to classical navigation interactive technique using joystick. This evaluation focused on the current acceptations of the realism concept [2]. A same driving task was proposed to subjects in three situations: in the real world with the targeted lift truck (R); in the virtual simulation of the scene in an immersive context using head tracking stereoscopy with a joystick interface to control the forklift (J); and in the same visual stereoscopic immersive environment but using the haptic driving paradigm we propose (H). Significant results are presented from analyses of subjects' performances, but also data on their behaviours, and their answers to subjective questionnaires. We discuss these results to put them in perspective, and then conclude.

2 Related work

Haptic devices have limited workspace, relative precision, and no complete range for rotational movements. In order to overcome these drawbacks, a basic interactive paradigm is the *clutching/declutching* technique which is however time consuming and which distracts the users' attention from the focus of their task. To avoid these drawbacks, several approaches use haptic devices to control navigation during interaction [3], using the relative position of an object of interest, the position of the haptic device, and the user position, to compute a more suitable point of view for the user to

interact with the object. Other solutions, such as the *Bubble* technique [4] or the *Haptic Hybrid Rotation* method [5], deal with the hardware limitations of the haptic device. These techniques provide a rate-control for object manipulation, based on an isomorphic mapping when the device is far from its workspace boundaries, and on a non-isomorphic mapping near the boundaries, also used by [6] or [7]. On the contrary to these methods our haptic paradigm is a non-isomorphic rate-control based on the comparisons of a tracked referential with a neutral one, such as [8]. This concept is augmented by an elastic force to retrieve this neutral referential, since it has been proved by [9] that such function is more suitable for rate control paradigms.

Some works also focus specifically on VR applications dedicated to forklifts. Safety is one of the application fields. [10] studied the impacts on the forklift occurring on drive-in racking structures. They proposed a general method for calculating forces generated under forklift truck impact. In [11] the prevention of forklift capsize is considered. Based on the fact that drivers' training and proper safety procedures are not sufficient enough to reduce accidents, they developed an intelligent control system, embedded on the forklift, which analyzes onboard sensors data and proposes some corrections. There are also some studies on real-time simulator with a forklift truck model, and on evaluation of numerical integrators to propose a simulation as realistic as possible [12]. Finally, [13] described a prototype of a full-immersive simulation of forklift truck operations for safety training.

Within this context, we propose below a haptic paradigm to provide sensorimotor stimulations which increase the realism of the interaction for tasks requiring physical involvement of the subjects.

3 System Description

This section describes first the physical behaviours of the virtual forklift. Then we detail the main features of the haptic driving paradigm itself.

3.1 Forklift physical behaviours

The inertia and damping of the vehicle in the real condition was obtained empirically, by velocities and deceleration measurements between two points. Actually, three forces are involved to move this kind of forklift: the weight of the vehicle (400 kg), its friction on the floor, and the force applied by the user to move it. We measured an empirically friction of wheels (stopping distance 4.25 m, initial velocity 1 m/s).

The realism of the interaction also supposes that the user has to feel the force on the forklift through the handle to make the virtual forklift move. In the one hand, this force is captured by the haptic arm. In the other hand, this is linked to the amplitude of the pressure exerted on the joystick. In both cases we need to express the new velocity of the vehicle.

Actually, acceleration of an object being the resulting force (F) applied on an object divided by its mass (m), while this acceleration being also the derivative of the velocity. We deduce from Eq.1 the expression of the new velocity according to F,

while Δt is the time variation, t the current time, V(t) the velocity at t, β the damping factor, m the object weight, and α a control constant.

$$V(t+\Delta t) = \left(V(t) e^{\frac{-\beta}{m}\Delta t} + \alpha \frac{F\Delta t}{m} \right) \quad (1)$$

All parameters of Eq. 1 are known, except β: this variable takes into account all the forces applied to the object, including the friction forces. In our physical simulation of the forklift we assume that β is constant, its variation being negligible. To determine its value, we ran the simulation with the forklift moving at a speed of 1m/s, and adapted the value of that constant to make this virtual vehicle stopping at the right distance (the one measured on the real forklift) from its starting point. The only parameter that one must define is α, a constant factor that we could define arbitrarily. That is actually useful, because we cannot apply on the haptic device the same force as the one used to move the real forklift.

In addition, we determined the curvature radius of the forklift, which only depends on the handle angle with the vehicle motion axis. We empirically find the linear mapping between the handle's angle and the inverse of the curvature radius. By this way we were able to compute the next position and orientation of the virtual forklift.

3.2 Haptic driving paradigm

Apart the rendering of the inertia and the damping of the forklift as external force feedbacks, our haptic driving paradigm simulates the mechanical behaviours of the real forklift's handle. Moreover, it provides an elegant solution for having an intuitive perception of the push and pull actions, while the base of the haptic device cannot move like the real forklift.

To simulate the real forklift's handle, we designed a virtual mechanism based on a rigid body with hinge constraints (Fig. 2 - right). A first hinge (blue sphere, red arrow) controls the orientation of the forklift's wheels. A second hinge (green arrow), allows user actions to raise or lower the forklift's handle

The haptic driving paradigm includes a push/pull haptic technique to control the velocity of the virtual forklift. This solution aims to provide a sensorimotor stimulation to the users during the driving task, by introducing a non-isomorphic rate-control of a virtual vehicle. However, on the contrary of the real truck, the base of the haptic device is fixed. Accordingly, we integrated a system of relative motion.

The blue sphere (Fig. 2 - right) is constraint along the green bar located under the blue sphere. This green bar follows the orientation's wheels that are control by the virtual handle. More the blue sphere goes forward, more the force applied by the user on the vehicle will be large and vice versa. Thus, the velocity of the virtual forklift is computed from the force applied by the user on the haptic devices, except it is constrained by the virtual mechanism. However, it is not possible to provide the same intensity of force feedback through the haptic device (device limitation) than in the

real context. That's why we applied a linear scaling on the forces computed by the physical simulation (cf. Eq. 1).

A principle for designing a relative motion system is to provide a non-isomorphic rate-control manipulation based on the position and angular distances with a neutral referential [8, 9]. In our case, it was sufficient to consider a neutral zone at the middle of the green bar, where we apply an elastic force which tends the blue sphere to return into this zone. Thus, more the user moves the sphere away from the neutral zone, more the strength to bring it back will be strong. Apart it provides the feeling of pushing or pulling harder, the user is navigating forward or backward with the virtual forklift in the full scene, in spite of moving only close to the device. A dead zone without elastic force around this neutral referential is also used, to avoid force oscillation and vibration, and to increase user comfort.

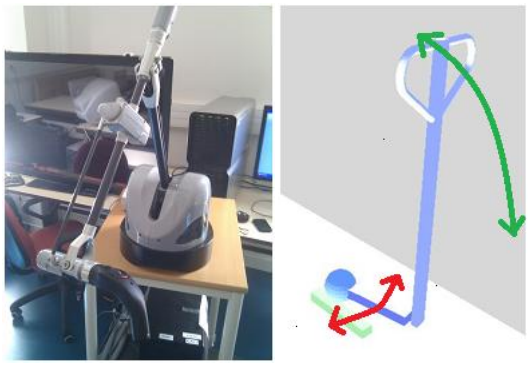


Fig. 2. The 6 DoF haptic arm that we used, a Virtuose 6D 35-45 of Haption (left). The virtual mechanism of the forklift handle (right) implemented with the Interactive Physical Pack (IPP) developed by Haption for the Virtools platform that we used for the visual immersion.

4 Evaluation

We evaluated a task of forklift driving in three experimental conditions: one with the real forklift (R), two others in a virtual context: with our haptic paradigm (H), and with a joystick interaction (J). Actually, a joystick can be considered as one of the most efficient interface to achieve navigation tasks in terms of completion time. However it doesn't provide sensorimotor feedbacks such as haptic devices. Thus, our fundamental questioning is the contribution of the sensorimotor feedbacks to the realism of the interaction. More specially, we hypothesis that our paradigm is better than joystick use for learning the driving task we wanted to address, because it makes possible a physical involvement of the user which is close to real situation. Accordingly, the physical rendering of the forklift collisions with the objects in the virtual scene is not our focus here. We mainly use the haptic device for simulating the motion constraints of the forklift handle, for rendering the inertia of this vehicle (non-motorized forklift), and for providing to the user a relative motion system. These three haptic feedbacks

provide a sensorimotor stimulation, which is supposed to allow a physical involvement of the user in the driving task.

4.1 Method

For the real condition (R), the non-motorized forklift was manipulated with its articulated handle in a 13x6 meters room (Figure 1, bottom). For the virtual context of the H and J conditions, we used a CAVE-like system, with adaptive stereoscopy based on the tracking of the subjects' head to provide the best visual immersion. Realistic 3D models of the real experimental room and the forklift have been designed (Figure 1, top). Moreover, the inertia behaviour on the virtual forklift was equally applied in H and J conditions. In this way, we can only focus on sensorimotor realism of our haptic paradigm, because the same level of visual realism was provided, whatever is the interactive device in the virtual context.

In the virtual conditions (H and J), we easily get the trajectory (Figure 3) of the forklift by logging the tracking events, the position and orientation of the virtual forklift. To get the same kind data in the real condition (R), all the trails were video captured, on which we applied classical image processing techniques based on the colour segmentation of the yellow and green spheres (top of the forklift - Figure 1), and a homographic projection to get a 2D trajectory on the floor. Using this data, we extracted a number of features on the driving tasks trajectories.

Concerning the subjective measures, we submitted to the users a questionnaire to analyze their perception of the task and their feedbacks. After each experimental condition, participants completed a set of rating-scale questions. Users had to report their level of agreement according to an eight-point Likert scale. We designed our questionnaire from different studies of presence [14, 15, 16]. Presence is a multifaceted concept, defined as the subjective experience of being in one place or environment, even when one is physically situated in another. We have set up a questionnaire based on four presence factors identified by [14], namely: control, sensory, realism, and distraction factors. We used the same questionnaire in the different conditions, with adaptations and deletions of some claims for the real world version.

18 subjects participated in the experiment: 14 males, 4 females, ranging between 24 and 57 years old (average = 35). All participants filled out a background questionnaire A large number of participants were people with a computer science background. Most of them have previously experienced interacting with 3D user interfaces (gamepad or joystick) in virtual environments (video games). Among these participants, 16 are right-handed and 2 are left-handed. Everyone used his dominant hand to operate with the proposed devices (real handle, joystick, haptic arm).

We asked orally to subjects to perform trajectories 8-shaped according to landmarks (traffic cones), 4 times starting from the right (Figure 3 - left), 4 times starting from the left (Figure 3 - centre). This driving task was performed after a training session based on a simpler driving task (strength line, left turn, right turn) to learn the manipulation of the real forklift, as well as the virtual one with the joystick or with the haptic paradigm. After the end of each experimental condition, participants completed

a set of rating-scale questions. Then they moved on to the next condition, following the same protocol, until all three conditions were executed.

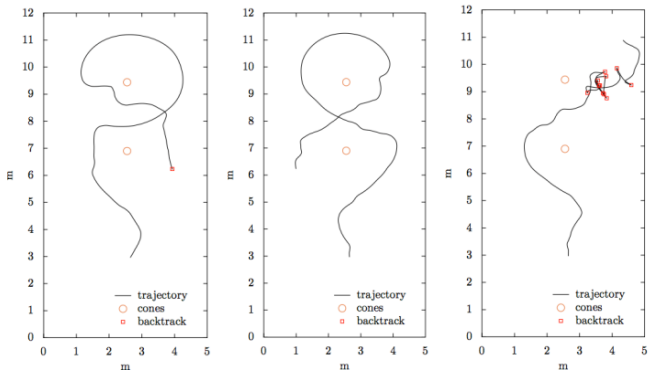


Fig. 3. Driving task through trajectory log analysis: success left path (left), success right path (centre), and a failed left path with a completion ratio of 50% and backtracks (right). Orange circles are the positions of landmarks (traffic cones), red rectangles represent the backtracks.

We performed a within-subject study (repeated measures) with the three forklift driving conditions. The execution order of the three conditions was randomized between subjects. Each subject performed the driving tasks successively without waiting time, except the time needed to complete the questionnaire after each condition. This setup was necessary for study the learning effects coming from the order of condition.

4.2 Performance analysis

We compared the number of failures and success of the driving task according to experimental conditions (R, J, H). This outcome of the task was analyzed with chi-square (χ^2) test. We observed that the driving task depended on the experimental conditions (χ^2 (2) = 22.44; $p < 0.00001$). Subjects failed the task more often in the virtual condition (J and H) than in the real one (R).

In order to compare experiment conditions, we had to separately analyze the trajectories features (time, distance, velocity, and backtrack) according to the success levels of the subjects. The results presented are statistically significant ($p<0.05$). Results are explicitly referred as a trend if p is between 0.05 and 0.1. We applied the Kolmogorov-Smirnov test to verify that the variables succeed in satisfying normality assumptions. We performed an ANOVA with experimental conditions as within-subject variables. Fisher's LSD was used for post-hoc pair-wise comparisons. All the analyses were performed with Statistica 9.

For subjects who succeed (S cluster), a number of the trajectories features depend of the experimental conditions. That is the case for: distances ($F(2, 162) = 8.03$; $p = 0.00047$), total duration of the task ($F(2, 162) = 23.551$; $p = 0.00001$), and number of backtracks ($F(2, 162) = 3.60$; $p = 0.029$). A post hoc analysis revealed that distances are shorter in R than in H and J. In addition, the task duration is faster in R and J than

in H (mean durations: 48 seconds for H, 30 seconds for R, and 28 seconds for J). And finally, a higher number of backtracks appeared for H, but no significant differences between J and R. However, all these results concern the S cluster, while for learning we are more interested by the subjects who failed.

For subjects who failed (F cluster), we observed that the distance varies according to the experimental conditions ($F(2, 114) = 7.65$; $p = 0.00076$). Before failing, the users covered a greater distance in R condition (20.55 m) than in virtual conditions (16.51m for J, 16.19m for H). In addition, the completion time varies according to the experimental conditions ($F(2, 114) = 13.41$; $p = 0.00001$). The mean duration of a driving task was 27 seconds for J condition whereas it was 43 seconds for R condition and 46 seconds for H condition. The users failed faster in J condition than in R and H conditions.

	Time	Distance	Velocity	Backtrack
S cluster				
HR	0.63	0.23	0.72	0.30
JR	0.39	ns	0.63	ns
HJ	0.35	ns	0.56	ns
F cluster				
HR	0.45	ns	0.76	ns
JR	ns	ns	ns	ns
HJ	ns	-0.33	ns	ns
All subjects				
HR	0.42	ns	0.78	0.22
JR	ns	ns	0.50	ns
HJ	0.27	ns	0.48	ns

Table 1. Set of correlations of the trajectories features between the different experimental conditions, for S and F clusters, and for all subjects.

Finally, we analyzed if the measures of the trajectories features were correlated in different conditions. The results show that for subjects who have succeed or failed (S and F clusters), and globally for all the subjects of the experiments, there is a strong or moderate correlation between H and R for completion time (blue - Table 1). We also notice that, except for subjects who have failed (F cluster), there is a weak correlation between H and R for backtracks (red - Table 1). Finally for velocity, the strongest correlations are observed between H and R (green - Table 1).

4.3 Subjects behaviour analysis

One acceptation of realism is the fact that behaviours or events observed in the real condition are reproduced in the virtual one [2]. To study the global behaviour of all subjects against condition task, we first focused on global standard deviation on curvilinear distance for each condition (H, J, R) when subject success 100% of the task, we obtain this relation: $SD_{Distance}(H)=5.05 > SD_{Distance}(R)=3.66 > SD_{Distance}(J)=2.19$.

Moreover, we looked at the curvilinear distance of the trajectories related to the 8 trails performed by each subject for each condition. For each condition, Figure 4 – left presents the standard deviation of the mean curvilinear distance of the 8 trajectories of the 18 subjects, while Figure 4 – right shows the mean of the 18 standard deviations of the curvilinear distances of the 8 trajectories. Then we studied the ranking of subject for each condition in terms of velocity and completion time. Figure 5 shows that the average difference of the variations of position compared to the ranking R, is twice larger for the ranking J than for the ranking H.

All together these results suggest that the subjects' behaviour in H condition is closer to the R condition than to the J one.

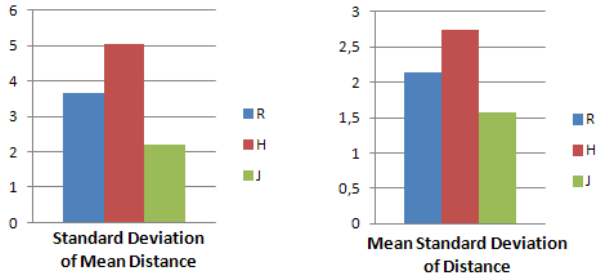


Fig. 4. The H condition increases inter-individual performance differences (left), as well as inter-trial performance of subjects (right), both also observed in the R condition.

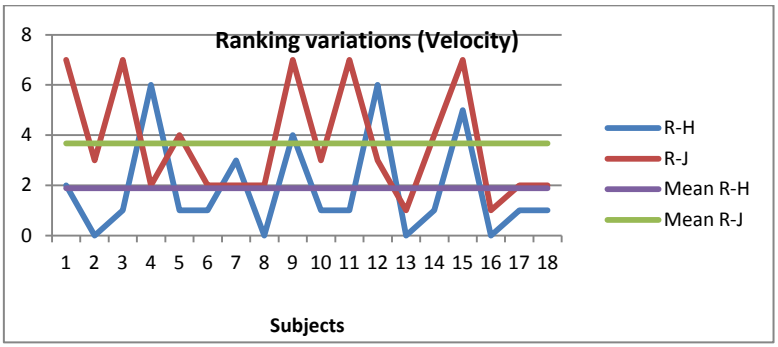


Fig. 5. Variations of subjects' positions in rankings of their velocities: 'Haptic' ranking is closer to 'Real' ranking than to the 'Joystick' one. The same occurs for the completion time.

4.4 Subjective data analysis

We performed an ANOVA with experimental conditions as within-subject variables. Fisher's LSD was used for post-hoc pair-wise comparisons.

On one hand, whatever is condition (R, H or J) no significant difference is observed concerning subjects' motivation for the task, while they judge that the interactive devices are all relatively intuitive. No difference is observed concerning a number of realism aspects or interface awareness: the subjects reported having lost conscious-

ness of the screens, not being affected by the visual display quality, not having the feeling of seeing a movie, and have seen many 3D objects. Concerning the sensory factor, the perception of the degree of movement was not affected by the experimental condition: the subjects said they were able to move in the scene, to change of perspective, and explore easy the scene in the real as well as in virtual conditions. All these results indicate that the virtual scene was realistic and well perceived by the subjects.

On the other hand, significant differences are observed for the control factor, and particularly for the mode of control. That is also interesting, because these features positively highlight natural aspect of the interaction with the environment. Firstly, the reported perception of external force feedbacks, of the inertia of the forklift, and of the physical implication varies according to experimental conditions (external force feedbacks: $F(2.34)=38.92$ $p<0.00001$; inertia $F(2.34)=21.56$ $p<0.0001$; physical implication: $F(2.34)=20.48$ $p<0.001$). Post hoc comparison showed that users perceive less external force feedbacks and inertia of the forklift, and being less physical implicated for the J condition than for the H and R conditions. Secondly, the perception of the behaviours of the forklift's handle varies according to the experimental condition ($F(2.34)=9.53$ $P<0.00005$). Post hoc comparison showed that subjects consider more that the forklift is manipulated by a handle in the R and H conditions than in the J condition. Thirdly, the J condition was rated as globally providing less natural interaction than the H condition. The J interaction was perceived as being less close of the R interaction than H interaction is closed to the R interaction. These results confirm our hypothesis. Lastly, the subjective measures show significant differences concerning the involvement in the forklift driving. The results suggest that the users are more focused on the manipulation paradigm in the R and H conditions than in the J condition. Indeed the subjects reported being more aware of the situation events ($F(2.34)=5.52$ $p<0.008$) and being more focused on the manipulation device ($F(2.34)=6.16$ $p<0.005$) in the R and H conditions than in the J condition.

5 Discussion

One of the definitions of realism - through the *action fidelity* or *functional fidelity* concepts [2], is that performance similarities between the virtual condition and real condition should correlate. It is especially the case for trajectories features (Table 1), which globally shows a better correlation of the performance between H and R than between J and R. On this field, the most important correlation between R and H is concerning velocity. Actually, except orientation control, velocity is the main parameter that user may control during the task. Velocity is a measure that characterizes a choice and strategy to succeed the task, and thus an individual specific behaviour. That means that in R and H conditions, the users adopt a same velocity strategy. This result corroborate with another observation: the user performance ranking on mean velocity in H condition is twice closer to the R condition ranking, than the J condition ranking is (Figure 5), which means that H seems to keep ranking on velocities.

The fact that there is the strongest correlation on velocities is an argument in favour of a better realism of H condition, according to [1] that supposes realism trans-

fers *behaviours* of the users between virtual and real condition. The way that users perform the task in the J condition is completely different than in the R condition. That also corroborates the subjects' answers to the questionnaires regarding the fidelity of the H condition with respect to the R one, when they declare that they have the feeling to manipulate the forklift handle in H condition, not in J.

Moreover the global standard deviation on curvilinear distance (Figure 4, left) highlights the fact that haptic condition reproduces and increases the high inter-individual differences observed in the R condition whereas these differences are blurred in the J condition. That is also a behaviours property, which is globally observed within the full group of subjects in real condition, and seems to be reproduced in haptic condition. In addition, at the individual scale, one must notices the weak homogeneity of performance between trajectories (Figure 4, right) performed by each user observed in R, was more observed in H than in J.

Another aspect of the realism is the learning ability that virtual environment may offer [1]. In that field we observed that when the users don't succeed the task, they have failed earlier in J condition than in H and R conditions. That is important because this forklift virtual simulator is basically dedicated for non-expert users that are supposed not really succeed the driving task when they are novices.

6 Conclusion

We designed an innovative driving paradigm for the virtual learning of forklifts, when a physical involvement of drivers is necessary in real context. On one hand, a rigid body model allows the users to perceive haptically the physical and mechanical behaviour of the forklift and its handle. On the other hand, the velocity control of the virtual forklift is simulated with a push/pull interactive technique. To evaluate this haptic paradigm, 18 subjects performed driving tasks in three conditions: with a non-motorized forklift within a real room, with virtual simulator of the forklift and the room, and for this second context, with a joystick interaction, and with our haptic paradigm. We covered the 6 conditions orders, and realised statistical analysis on trajectories performances, subjects' behaviours, and on subjective data.

One dimension of realism supposes that a performance transfer must be observed between real and virtual contexts. On this acceptation, we basically found some interesting correlations between real driving condition and our haptic driving paradigm, mainly for completion time and velocity. But our haptic technique clearly appeared more realistic than the joystick one, when considering another dimension of realism, namely the behaviour and psychological processes transfers. Actually, real and haptic conditions are correlated on several behaviours: velocity chosen by the subject, conservation of subjects' rankings for velocity, completion time, and behaviours at individual and group scales (inter-individual and inter-trial performance differences). This dimension of the realism in favour of our haptic paradigm has been also highlighted within the subjective data analysis: good perception of external forces and inertia of the forklift, feeling to be physically involved, feeling to manipulate the forklift by a handle, better focus on the driving task. Accordingly, we may globally conclude that

our haptic paradigm provides more realism than the joystick interaction, for the virtual driving of such kind of forklifts.

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